WORKSHEET — UNIT III

Quantum Algorithms

Course: Fundamentals of Quantum Computing

Unit: III (5 Hrs / 300 min)

Topics: What Makes an Algorithm Quantum; Oracles and Quantum Parallelism; Deutsch and Deutsch–Jozsa Algorithms; The Quantum Fourier Transform; Grover's Search Algorithm; Shor's Factoring Algorithm; Speedup Analysis and the Limits of Quantum Advantage

🎯 Section A — Learning Objectives

1. Describe the general recipe of a quantum algorithm (encode, superpose, apply oracle, interfere, measure) and distinguish query complexity from time complexity.

2. Build a reversible oracle Uf, explain quantum parallelism and why it is not enough alone, and use phase kickback.

3. Trace the Deutsch, Deutsch–Jozsa and Bernstein–Vazirani algorithms and compare their query counts with classical methods.

4. Define the quantum Fourier transform, compute a 2-qubit QFT by hand, and explain phase estimation and the $O(n^2)$ gate count.

5. Explain Grover's search: the phase oracle, the diffusion operator, the geometric picture, the optimal number of iterations and over-rotation.

6. Explain Shor's algorithm (order finding, factoring 15) and judge the limits of quantum advantage using speedups and the classes P, NP and BQP.

🔑 Section B — Key Concepts

No.	Concept	Short Explanation
1	**General Recipe**	Most quantum algorithms encode the input, create a superposition, apply an oracle, use interference to boost the right answer, then measure.
2	**Query vs Time Complexity**	Query complexity counts calls to the oracle; time complexity counts every elementary operation of the whole algorithm.
3	**Oracle and Quantum Parallelism**	The oracle Uf maps $\|x\rangle\|y\rangle$ to $\|x\rangle\|y \oplus f(x)\rangle$ and can act on a superposition of all inputs at once, but a measurement returns only one result.
4	**Phase Kickback**	With the target in $\|-\rangle$, the oracle leaves the target unchanged and multiplies the input by $(-1)^{f(x)}$, turning outputs into phases.
5	**Deutsch and Deutsch–Jozsa**	One quantum query decides whether f is constant or balanced; a classical deterministic method needs $2^{n-1} + 1$ queries in the worst case.
6	**Bernstein–Vazirani**	One quantum query finds the hidden bit string s in $f(x) = s \cdot x \pmod 2$, where a classical method needs n queries.
7	**Quantum Fourier Transform**	The QFT maps $\|j\rangle$ to an equal superposition with phases $e^{2\pi ijk/N}$; it needs only $O(n^2)$ gates for n qubits.
8	**Quantum Phase Estimation**	It uses controlled-U operations and the inverse QFT to read out the eigenphase φ of a unitary U.
9	**Grover's Search**	An oracle marks the solution, the diffusion operator inverts about the mean, and about $(\pi/4)\sqrt{N}$ iterations find the item with high probability.
10	**Shor's Algorithm**	Factoring reduces to finding the period of $a^x \bmod N$; the quantum part finds the period with the QFT and a classical part extracts the factors.
11	**Limits of Advantage**	Speedups may be quadratic (Grover) or superpolynomial (Shor); BQP is not known to contain NP-complete problems.

📘 Section C — Important Terminology

Keyword	Meaning
Query Complexity	The number of calls an algorithm makes to the oracle.
Time Complexity	The total number of elementary steps an algorithm needs.
Oracle (Black Box)	A circuit Uf that evaluates a function f whose inner workings are hidden from the algorithm.
Quantum Parallelism	Applying a function to all basis states of a superposition in a single oracle call.
Phase Kickback	The effect by which the oracle's output appears as a phase $(-1)^{f(x)}$ on the input register.
Constant Function	A function that returns the same output for every input.
Balanced Function	A function that returns 0 for exactly half of the inputs and 1 for the other half.
Deutsch–Jozsa Algorithm	An algorithm that decides constant versus balanced with a single oracle query.
Bernstein–Vazirani Algorithm	An algorithm that finds a hidden string s with a single query to $f(x) = s \cdot x \bmod 2$.
Discrete Fourier Transform	A classical transform that converts N values into N frequency components.
Quantum Fourier Transform	The quantum version of the DFT, mapping $\|j\rangle$ to $(1/\sqrt{N}) \Sigma e^{2\pi ijk/N}\|k\rangle$.
Controlled-Phase Rotation	A two-qubit gate that applies a phase rotation to the target when the control is 1; it is a building block of the QFT.
Quantum Phase Estimation	A procedure that estimates the eigenphase φ in $U\|u\rangle = e^{2\pi i\varphi}\|u\rangle$.
Phase Oracle	An oracle that flips the sign of the marked state(s) and leaves all other states unchanged.
Diffusion Operator	Grover's "inversion about the mean" operator, $2\|s\rangle\langle s\| - I$, which amplifies the marked amplitude.
Amplitude Amplification	The general framework, of which Grover's search is a special case, for boosting the amplitude of desired states.
Over-Rotation	Applying too many Grover iterations, which rotates the state past the target and lowers the success probability.
Order (Period) Finding	Finding the smallest $r > 0$ such that $a^r \equiv 1 \pmod N$.
Modular Exponentiation	Computing $a^x \bmod N$, the function whose period Shor's algorithm finds.
Continued Fractions	A classical method used to recover the period from a measured phase estimate.
RSA	A public-key cryptosystem whose security rests on the difficulty of factoring a large integer.
BQP	The class of decision problems a quantum computer can solve in polynomial time with bounded error.
P and NP	P: problems solvable in polynomial time by a classical computer; NP: problems whose solutions can be checked in polynomial time.
Quantum Advantage	A task on which a quantum device outperforms the best known classical methods.

✅ Section D — Multiple-Choice Questions [Easy–Medium]

1. Which sequence matches the general recipe of a quantum algorithm?

A. Measure → encode → oracle → superpose → interfere
B. Encode → superpose → apply oracle → interfere → measure
C. Oracle → measure → encode → interfere → superpose
D. Superpose → measure → oracle → encode → interfere

2. What does the oracle Uf do to the state $|x\rangle|y\rangle$?

A. It maps it to $|x \oplus y\rangle|f(x)\rangle$
B. It maps it to $|f(x)\rangle|y\rangle$
C. It maps it to $|x\rangle|y \cdot f(x)\rangle$
D. It maps it to $|x\rangle|y \oplus f(x)\rangle$

3. With the target qubit in $|-\rangle$, what does Uf do to $|x\rangle|-\rangle$ (phase kickback)?

A. It gives $(-1)^{f(x)}\ |x\rangle|-\rangle$
B. It gives $|x\rangle|+\rangle$
C. It gives $f(x)\ |x\rangle|-\rangle$
D. It gives $|f(x)\rangle|-\rangle$

4. How many oracle queries does Deutsch's algorithm need to decide whether f is constant or balanced?

A. 0
B. 2
C. 1
D. 4

5. What is the classical deterministic worst case for the n-bit Deutsch–Jozsa problem?

A. 2^n queries
B. $2^{n-1} + 1$ queries
C. n queries
D. 1 query

6. In the Deutsch–Jozsa algorithm, a final measurement of all zeros means that f is …

A. balanced
B. random
C. faulty
D. constant

7. What does the Bernstein–Vazirani algorithm find with one quantum query?

A. The hidden string s
B. The period of a function
C. The marked item in a database
D. A factor of an integer

8. How many gates does the QFT on n qubits need?

A. $O(2^n)$
B. O(N log N)
C. $O(n^2)$
D. $O(n\ 2^n)$

9. How many Grover iterations are about optimal for N = 16 with one marked item?

A. 1
B. 3
C. 8
D. 16

10. Which statement about Shor's algorithm is correct?

A. It uses Grover search to test every possible factor
B. It needs no classical post-processing
C. It takes exponential time on a quantum computer
D. It reduces factoring to finding the period of $a^x \bmod N$

✏️ Section E — Fill in the Blanks [Easy]

1. In Deutsch's problem, the function f is either constant or _____.

2. The effect in which the oracle's output appears as a phase (−1)^{f(x)} on the input register is called phase _____.

3. Grover's search needs about √N queries instead of N, which is a _____ speedup.

4. Shor's algorithm reduces factoring to finding the _____ (order) of $a^x \bmod N$.

5. The class of problems a quantum computer can solve in polynomial time with bounded error is called _____.

☑️ Section F — True or False [Easy]

1. Evaluating f on a superposition of all inputs lets us read out every value of f directly. (True / False)

2. The Bernstein–Vazirani algorithm needs one quantum query, whereas a classical method needs n queries. (True / False)

3. In Grover's algorithm, more iterations always give a higher chance of finding the marked item. (True / False)

4. The quantum part of Shor's algorithm uses the QFT to find a period. (True / False)

5. Quantum computers are known to solve NP-complete problems in polynomial time. (True / False)

🔗 Section G — Match the Following [Medium]

Column A	Column B (shuffled)
1. Oracle	a. Finding the smallest $r > 0$ with $a^r \equiv 1 \pmod N$, the core of Shor's algorithm
2. Phase kickback	b. Inversion about the mean, which amplifies the marked amplitude
3. Deutsch–Jozsa algorithm	c. Black-box circuit that computes f reversibly
4. Quantum Fourier transform	d. Decides constant versus balanced with a single query
5. Diffusion operator	e. Maps |j⟩ to an equal superposition with phases e^{2πijk/N}
6. Order finding	f. Turns the oracle's bit flip on |−⟩ into a phase on the input register
7. BQP	g. Problems solvable by a quantum computer in polynomial time with bounded error

📝 Section H — Short-Answer Questions [Medium]

1. State the general recipe of a quantum algorithm and explain the difference between query complexity and time complexity.

2. What is phase kickback? Show what Uf does to |x⟩|−⟩.

3. Explain how Deutsch's algorithm decides in a single query whether f is constant or balanced, and say what a classical method needs.

4. For N = 16 with one marked item, compare the classical and Grover query counts and give the optimal number of Grover iterations.

5. Why does Shor's algorithm threaten RSA, and which part of it is quantum?

📄 Section I — Descriptive Questions [Medium–Advanced]

1. Explain the oracle model and quantum parallelism, building a reversible oracle for a Boolean function and using phase kickback, then trace the Deutsch, Deutsch–Jozsa and Bernstein–Vazirani algorithms, including a 2-bit worked example.

2. Describe the quantum Fourier transform and quantum phase estimation: the definition, the circuit of Hadamards and controlled-phase rotations, a 2-qubit QFT by hand, the inverse QFT, and the $O(n^2)$ versus $O(N \log N)$ comparison.

3. Explain Grover's search: the phase oracle, the diffusion operator, the geometric picture, the optimal number of iterations, over-rotation, the N = 16 comparison and amplitude amplification.

4. Discuss Shor's algorithm (reduction to order finding, classical and quantum parts, factoring 15, resource estimates for RSA-2048) and then analyse the limits of quantum advantage using quadratic versus exponential speedups, P, NP and BQP, and the quantum-advantage experiments.

💻 Section J — Practical / Application-Based Questions [Advanced]

Practical 1

Problem: Compute Grover's success probability for each number of iterations.

Task:

1. Define grover_prob(N, k) = $\sin^2((2k + 1)\cdot\theta)$, where $\theta = \text{asin}(1/\sqrt{N})$, for one marked item among N.

2. For N = 16, print the probability for k = 0, 1, 2, 3 and 4 rounded to 4 decimal places.

3. Print the value of k that gives the highest probability.

Expected Output: *Six lines: 0.0625, 0.4727, 0.9084, 0.9613, 0.5817, then best k = 3.*

Practical 2

Problem: Write the classical post-processing step of Shor's algorithm.

Task:

1. Define shor_classical(N, a) that finds the order r, the smallest r > 0 with a^r mod N = 1.

2. Return (r, gcd(a^(r/2) − 1, N), gcd(a^(r/2) + 1, N)) when r is even.

3. Test it with N = 15 and a = 7, a = 2 and a = 4, and print each result.

Expected Output: *Three lines: (4, 3, 5), (4, 3, 5), (2, 3, 5).*

Practical 3

Problem: Simulate Deutsch's algorithm for four one-bit functions.

Task:

1. Represent the two-qubit state as a list of 4 amplitudes (index 2x + y) and start from $|01\rangle$.

2. Apply H to both qubits, then the oracle U_f: $|x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$, then H to the first qubit.

3. For f(x) = 0, f(x) = 1, f(x) = x and f(x) = 1 − x, print the probability that the first qubit reads 1.

Expected Output: *Four lines: 0.0, 0.0, 1.0, 1.0 (0 means constant, 1 means balanced).*

🗺️ Section K — Concept-Mapping Activity

QUANTUM ALGORITHMS

Complete the map below: fill in the missing sub-topic name or example for each branch.

Branch 1	Branch 2	Branch 3	Branch 4
Oracle-Based Algorithms	**Fourier-Based Algorithms**	**Search Algorithms**	**Limits and Complexity**

Example: _____	Example: _____	Example: _____	Example: _____

(Answer key — example values: Deutsch–Jozsa, Bernstein–Vazirani, phase kickback; QFT, phase estimation, Shor's algorithm; Grover's search, diffusion operator, amplitude amplification; BQP, NP-complete problems, quantum-advantage experiments)

🔬 Section L — Case Study / Real-World Activity

Scenario

A bank's security team is judging the long-term threat of quantum computers. Its web servers use RSA-2048 for key exchange, its archives use AES-128 encryption, and one analyst proposes using Grover's algorithm to search an unsorted table of 2^{20} records for a single matching entry. A vendor also claims that its new device "solves NP-complete problems exponentially faster".

Questions

1. For the table of $N = 2^{20}$ records with one match, estimate the classical average query count and the number of Grover iterations.

2. Which algorithm threatens RSA-2048, and which part of it is quantum? What does Grover's algorithm do to the effective strength of AES-128, and how can the bank restore its margin?

3. Explain why Shor's speedup is a bigger threat than Grover's, and comment on the vendor's claim using the classes P, NP and BQP.

4. Give two reasons why claims of quantum advantage are debated.

Expected Learning Outcome

Students apply quantum-algorithm speedups and complexity classes to judge realistic security risks and to examine claims of quantum advantage critically.

📊 Section M — Self-Assessment

☐ **Yes** ☐ **Partly** ☐ **No** I can explain the major concepts of this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can define the important terminology from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can solve problems/exercises from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can apply the concepts of this unit to a new situation.

💭 Section N — Student Reflection

What did I learn in this unit?

What was difficult for me?

What do I need to practice more?

One important concept I learned:

🔑 Section O — Instructor Answer Key

(For instructor use — covers all objective and subjective items in this worksheet.)

MCQ Answers

- 1. B — Encode the input, create a superposition, apply the oracle, let amplitudes interfere, then measure.

- 2. D — The oracle XORs f(x) into the target register and leaves the input register unchanged, which makes it reversible.
- 3. A — With $|-\rangle$ in the target, the bit flip becomes a sign: $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$.
- 4. C — One query on a superposition of inputs is enough for Deutsch's algorithm.
- 5. B — A deterministic classical algorithm may need to see more than half of the 2^n inputs, i.e. $2^{n-1} + 1$ queries.
- 6. D — The all-zeros outcome has amplitude ±1 only if f is constant; for a balanced f its amplitude is 0.
- 7. A — One query yields the hidden string s; a classical method needs n queries.
- 8. C — The QFT circuit uses n Hadamards and about $n^2/2$ controlled-phase gates, i.e. $O(n^2)$.
- 9. B — $(\pi/4)\sqrt{16} = \pi \approx 3.14$, so 3 iterations (success probability about 0.96).
- 10. D — Shor's algorithm finds the period of $a^x \bmod N$ with the QFT and then derives factors classically with gcd.

Fill in the Blanks — Answers

- 1. balanced
- 2. kickback
- 3. quadratic
- 4. period
- 5. BQP

True/False — Answers

- 1. False
- 2. True
- 3. False
- 4. True
- 5. False

Match the Following — Answer Key

1 → c 2 → f 3 → d 4 → e 5 → b 6 → a 7 → g

Short Answer — Key Points

- 1. Recipe: encode → superpose → apply oracle → interfere → measure. Query complexity counts oracle calls only, whereas time complexity counts all elementary operations, including the gates around the oracle.
- 2. Phase kickback: when the target is $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, the XOR with f(x) multiplies the target by $(-1)^{f(x)}$; this factor can be moved to the input, so $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$.
- 3. Start with $|0\rangle|1\rangle$ and apply H to both qubits; the oracle then puts $(-1)^{f(x)}$ on each term of the first qubit. A final H on the first qubit gives $|0\rangle$ if f is constant and $|1\rangle$ if f is balanced. A classical method needs two queries.
- 4. Classical: about N/2 = 8 queries on average, 16 in the worst case. Grover: about $(\pi/4)\sqrt{16} \approx 3$ iterations, with success probability about 0.96.
- 5. RSA relies on the difficulty of factoring. Shor's algorithm factors in polynomial time: the quantum part (period finding with the QFT) is the speedup, while the gcd and continued-fraction steps are classical.

Descriptive Questions — Key Points

- 1. Oracle: $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$. Parallelism: U_f on $\Sigma|x\rangle$ evaluates f on all inputs, but measuring gives one result, so interference is needed. Phase kickback turns outputs into phases. Example oracle: f(x1, x2) = x1 AND x2 is a Toffoli gate. Deutsch: H on both qubits, U_f, H on the first qubit, measure (0 = constant, 1 = balanced). DJ for n bits: n Hadamards, U_f, n Hadamards; all zeros means constant. 2-bit trace: with a balanced f, the amplitude of $|00\rangle$ is 0. Classical worst case for n = 2: 2 + 1 = 3 queries versus 1. BV: same circuit, the output is s.
- 2. $QFT|j\rangle = (1/\sqrt{N}) \Sigma_k e^{2\pi ijk/N}|k\rangle$. Circuit: Hadamard and controlled-phase rotations on each qubit, then swaps. For N = 4, $QFT|1\rangle = (1/2)(|0\rangle + i|1\rangle - |2\rangle - i|3\rangle)$. The inverse QFT reverses the circuit. Phase estimation: Hadamards on counting qubits, controlled-U^{2^j}, then inverse QFT, which gives the eigenphase φ in binary. Gates: $O(n^2)$ versus classical FFT $O(N \log N) = O(n\, 2^n)$.
- 3. Phase oracle flips the sign of the solution; diffusion $(2|s\rangle\langle s| - I)$ inverts the amplitudes about their mean; together they rotate the state by 2θ in a plane spanned by the solution and the rest, with $\sin\theta = 1/\sqrt{N}$. Optimal $k \approx (\pi/4)\sqrt{N}$. Too many

iterations over-rotate and lower the probability (for N = 16, k = 3 gives 0.96 but k = 4 gives 0.58). Classical N = 16: about 8 queries on average versus 3 for Grover. Amplitude amplification generalizes this to any procedure that has some success amplitude.

- 4. Shor: factoring reduces to finding the order r of a mod N; classical steps are gcd, modular exponentiation and continued fractions; the quantum step is period finding with the QFT on a register of about 2n qubits plus an n-qubit work register. Factoring 15 with a = 7: the order is 4, since $7^4 = 2401 \equiv 1 \pmod{15}$; $\gcd(7^2 - 1, 15) = \gcd(48, 15) = 3$ and $\gcd(7^2 + 1, 15) = \gcd(50, 15) = 5$. Resource estimates for RSA-2048 have fallen from tens of millions of noisy physical qubits to under a million in recent estimates and keep changing. Grover is quadratic, Shor is superpolynomial. $P \subseteq BQP \subseteq PSPACE$; NP-complete problems are not believed to be in BQP; factoring is in BQP but not known to be in P. Many problems have no known advantage. Advantage experiments (e.g. random-circuit sampling) are debated because improved classical simulations narrow the gaps.

Practical Activities — Expected Outputs

- 1. Sample logic: theta = math.asin(1/math.sqrt(N)); return math.sin((2*k + 1)*theta)**2. For N = 16 the values are 0.0625, 0.4727, 0.9084, 0.9613 and 0.5817; the best k is 3.
- 2. Sample logic: r = 1; while pow(a, r, N) != 1: r += 1; then x = a**(r//2) and return (r, math.gcd(x − 1, N), math.gcd(x + 1, N)). The outputs are (4, 3, 5), (4, 3, 5) and (2, 3, 5).
- 3. Sample logic: index i = 2x + y; H on a qubit mixes pairs of amplitudes with $\pm 1/\sqrt{2}$; Uf moves amplitude from (x, y) to (x, y ⊕ f(x)). The probability that the first qubit is 1 is the sum of the amplitudes squared for indices 2 and 3: 0.0 and 0.0 for the constant functions and 1.0 and 1.0 for the balanced ones.

Concept Map — Sample Answers

- Oracle-Based Algorithms: Deutsch, Deutsch–Jozsa, Bernstein–Vazirani, phase kickback. Fourier-Based Algorithms: QFT, phase estimation, Shor’s algorithm. Search Algorithms: Grover’s search, phase oracle, diffusion operator, amplitude amplification. Limits and Complexity: P, NP, BQP, quadratic versus exponential speedup, quantum-advantage experiments.

Case Study — Key Points

- 1. Classical average ≈ N/2 = 524,288 queries; Grover ≈ $(\pi/4)\sqrt{(2^{20})} = (\pi/4) \cdot 1024 \approx 804$ iterations.
- 2. Shor’s algorithm threatens RSA-2048; the period finding with the QFT is the quantum part. Grover gives only a quadratic speedup, so AES-128 behaves like a roughly 64-bit key against a quantum search; moving to AES-256 restores the margin.
- 3. Shor gives a superpolynomial speedup that breaks RSA outright, while Grover’s quadratic speedup can be offset by longer keys. NP-complete problems are not known to be in BQP, so the vendor’s claim would need strong evidence; known speedups for them are at best quadratic.
- 4. Possible reasons: improved classical simulations can match the claimed quantum advantage, current devices are noisy with few qubits, the benchmark tasks have little practical use, and the quantum speedup is often measured against weak classical baselines.